CSE STUDY HUB

Digital Forensics & Incident Response

UNIT 3 · System and Network Forensics

What this unit is about

This unit builds understanding of **System and Network Forensics**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. File systems, partitions, metadata, timestamps and artefacts

2. Deleted-file recovery, unallocated space, slack space and file carving

3. Operating-system artefacts: event logs, registry/configuration, user activity and services

4. Browser history, downloads, cookies and email artefacts

5. Memory forensics: processes, connections and volatile artefacts

6. Network evidence: packet capture, flow records, DNS, firewall and proxy logs

7. Timeline reconstruction and correlating evidence sources

8. Data recovery: logical vs physical recovery; effects of overwriting

9. Forensic tools, validation, limitations and reporting

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 3 NOTES · PART 1

File systems, partitions, metadata, timestamps and artefacts

File systems, partitions, metadata, timestamps and artefacts is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Deleted-file recovery, unallocated space, slack space and file carving

Deleted-file recovery, unallocated space, slack space and file carving is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Operating-system artefacts: event logs, registry/configuration, user activity and services

Operating-system artefacts: event logs, registry/configuration, user activity and services is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on System and Network Forensics, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 2

Browser history, downloads, cookies and email artefacts

A professional email has a specific subject, suitable greeting, purpose-first opening, concise body, clear action/deadline, polite close and signature. Check tone, recipients and attachments before sending.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Memory forensics: processes, connections and volatile artefacts

Volatile evidence can disappear when power is lost or a system changes. RAM, running processes, active network connections and logged-in sessions should generally be considered before persistent disk data.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Network evidence: packet capture, flow records, DNS, firewall and proxy logs

Network evidence: packet capture, flow records, DNS, firewall and proxy logs is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on System and Network Forensics, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 3 NOTES · PART 3

Timeline reconstruction and correlating evidence sources

Timeline reconstruction and correlating evidence sources is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Data recovery: logical vs physical recovery; effects of overwriting

Compare the alternatives in Data recovery: logical vs physical recovery; effects of overwriting using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Forensic tools, validation, limitations and reporting

Compare the alternatives in Forensic tools, validation, limitations and reporting using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on System and Network Forensics, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **File systems, partitions, metadata, timestamps and artefacts** — know the definition, core working and one example.

• **Deleted-file recovery, unallocated space, slack space and file carving** — know the definition, core working and one example.

• **Operating-system artefacts: event logs, registry/configuration, user activity and services** — know the definition, core working and one example.

• **Browser history, downloads, cookies and email artefacts** — know the definition, core working and one example.

• **Memory forensics: processes, connections and volatile artefacts** — know the definition, core working and one example.

• **Network evidence: packet capture, flow records, DNS, firewall and proxy logs** — know the definition, core working and one example.

• **Timeline reconstruction and correlating evidence sources** — know the definition, core working and one example.

• **Data recovery: logical vs physical recovery; effects of overwriting** — know the definition, core working and one example.

• **Forensic tools, validation, limitations and reporting** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from System and Network Forensics. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define File systems, partitions, metadata, timestamps and artefacts and explain its importance.

Q2. Explain the working/steps involved in Deleted-file recovery, unallocated space, slack space and file carving.

Q3. Differentiate between two related concepts from System and Network Forensics.

Q4. Give one practical application of Forensic tools, validation, limitations and reporting and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of System and Network Forensics.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.